

Iron sheath and dagger
from Mesopotamia
(now modern Iraq)



Iron and steel

Although iron was smelted as early as 2500 BCE, it was later discovered that heating it with a carbon material such as charcoal at a higher temperature resulted in a much harder metal. This strengthened iron, or steel, became common around 1200 BCE in

Anatolia (present-day Turkey). The new process allowed the production of stronger weapons and tools.

“There is a mine for silver and a place where gold is refined. Iron is taken from the earth and copper is smelted from ore.”

Bible, Book of Job, Chapter 28, verses 1–2



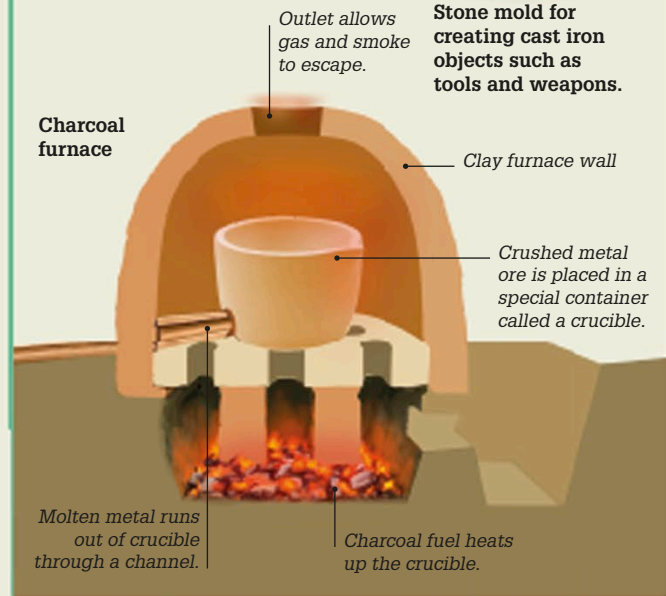
Molten copper flows
through a channel
and is collected.

Casting

The first furnaces produced a spongy mass of iron containing impurities that had to be hammered out. Around 900 BCE, in China, furnaces were developed that heated the iron ore up to a higher temperature to produce only pure iron. The molten metal was poured, or cast, directly into molds to make objects.



Stone mold for
creating cast iron
objects such as
tools and weapons.



Charcoal
furnace

Outlet allows
gas and smoke
to escape.

Clay furnace wall

Crushed metal
ore is placed in a
special container
called a crucible.

Molten metal runs
out of crucible
through a channel.

Charcoal fuel heats
up the crucible.



Gilded Roman
necklace with
semi-precious
stones, c 1st
century CE

Gilding

The art of gilding, or covering objects with a fine layer of gold leaf, was carried out as early as 3000 BCE.

In the 1st century CE, Roman goldsmiths began to make amalgam, a fine paste of mercury and gold, which stuck better to the surface it was coating.



Egyptian mirror
made of copper

c 1400 BCE

Pewter, an alloy of copper, antimony, and lead, was first produced in the Middle East. It was often used for vessels and tableware.

c 1300 BCE

Metalworkers added carbon to iron when smelting. This produced steel, a much stronger form of iron.

900 BCE

The process of producing cast iron was discovered in China. Using this technique, metal objects were created by pouring molten iron into molds.

c 100 CE

Roman metalworkers created amalgam, a mix of mercury and gold that made a more durable material for gilding than gold leaf.